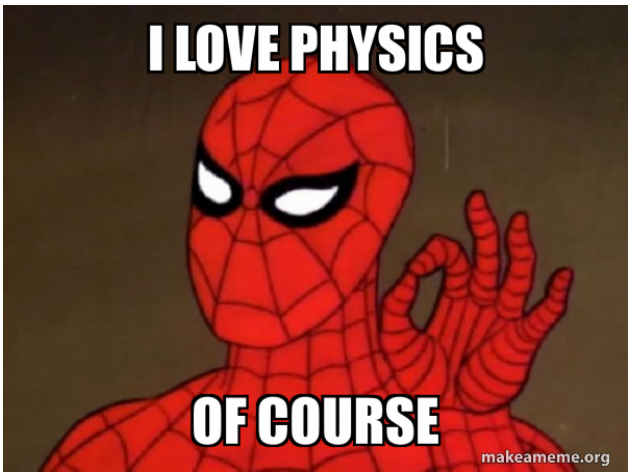

Grade 12 Physics
Final Exam Key

First Name: _____**KEY**_____

Last Name: _____

Directions:

- Please answer to 2 decimal points
- The test is designed to be completed in 75 minutes

For grading use only

Page:	2	3	4	5	Total
Points:	7	8	13	27	55
Score:					

Multiple Choice (10 marks)

1. (1 point) A projectile launched at 30° has range R. If launched at 60° with same speed, its range will be:
   - A. $\frac{R}{2}$
   - **B.** R
   - C. $\frac{2R}{3}$
   - D. $2R$

2. (1 point) The work done by conservative forces equals:
   - **A. Negative change in potential energy**
   - B. Total mechanical energy
   - C. Change in kinetic energy
   - D. Power multiplied by time

3. (1 point) A projectile is launched at 50 m/s at 60°. Its maximum height is:
   - **A.** 96 m
   - B. 100 m
   - C. 125 m
   - D. 150 m

4. (1 point) The area under an acceleration-time graph represents:
   - A. Displacement
   - **B. Change in velocity**
   - C. Total distance
   - D. Kinetic energy

5. (1 point) When a ball hits a wall and bounces back, the forces involved:
   - A. Depend on the ball's material
   - B. Are greater on the wall
   - C. Cancel each other completely
   - **D. Are equal in magnitude and opposite in direction**

6. (1 point) A car rounds a curve with radius r at speed v. If speed doubles, required centripetal force becomes:
   - A. 2F
   - **B. 4F**
   - C. F/2
   - D. F/4

7. (1 point) A 10 kg box slides 5 m with =0.4. Work done by friction is:
   - A. 200 J
   - B. 0 J
   - **C.** −196 J
   - D. 392 J

8. (1 point) Mechanical energy is conserved in:
   A. A swinging pendulum with air resistance
   B. A roller coaster descending a frictionless track
   C. A car braking to stop
   D. A bouncing ball

9. (1 point) A 5 kg object changes velocity from $-5\,\mathrm{m/s}$ to $10\,\mathrm{m/s}$. Impulse is:
   A. 75 Ns
   B. 25 Ns
   C. -25 Ns
   D. 50 Ns

10. (1 point) A 2000 kg truck (20 m/s) collides with 1500 kg car at rest. Final speed if they stick:
    A. 20 m/s
    B. 15 m/s
    C. 8.6 m/s
    D. 11.4 m/s

11. (1 point) Gravitational field strength on planet with 2M and 2R compared to Earth:
    A. 2g
    B. g
    C. 0.25g
    D. 0.5g

12. (1 point) A 4 µC charge moves 2 m in 300 N/C field. Work done by field:
    A. 2.4×10^{-3} J
    B. 2.4×10^{-3} J
    C. -2.4×10^{-3} J
    D. 4.8×10^{-3} J

13. (1 point) Light enters water (n=1.33) from air at 30°. Refracted angle:
    A. 22°
    B. 30°
    C. 22°
    D. 48.6°

14. (1 point) A diffraction grating with 1000 lines/mm shows 2nd order for 500 nm at:
    A. No solution
    B. 30°
    C. 60°
    D. 90°

15. (1 point) A spaceship moving at 0.8c experiences 10 years. Earth observers measure:
    A. 16.7 years
    B. 6 years
    C. 10 years

D. 8 years

16. (1 point) Metal with work function 4.0 eV has threshold wavelength:
    A. 300 nm
    B. 400 nm
    C. 310 nm
    D. 620 nm

17. (1 point) Plane flies 250 m/s north with 50 m/s west wind. Ground speed:
    A. 200 m/s
    B. 255 m/s
    C. 250 m/s
    D. 245 m/s

18. (1 point) Equal masses in Atwood machine. Tension is:
    A. Zero
    B. mg
    C. 2mg
    D. mg

19. (1 point) 60 kg student climbs 5 m in 10 s. Average power:
    A. 300 W
    B. 150 W
    C. 294 W
    D. 600 W

20. (1 point) 0.5 kg east at 10 m/s + 0.8 kg north at 8 m/s. Final speed if stuck:
    A. 9.0 m/s
    B. 8.1 m/s
    C. 12.4 m/s
    D. 6.5 m/s

Short Answer (35 marks)

21. **Football Field Goal Attempt:** During a crucial game, the kicker launches the football at 40 m/s from ground level at a 60° angle. The crossbar is 3 m high and 45 m horizontally from the kick point.
    (a) (3 points) Calculate the maximum height reached by the football
    (b) (2 points) Determine if this kick would result in a successful field goal

Solution:

22. **Mountain Rescue:** A 1200 kg rescue helicopter hovers stationary while lowering a paramedic package using a cable. The package (85 kg) accelerates downward at $2\,\mathrm{m/s^2}$.
    (a) (3 points) Calculate the tension in the cable during descent

(b) (2 points) Determine the upward force the helicopter must maintain

Solution:

23. **Warehouse Loading Ramp Safety:** Workers need to push a 8 kg package up a 35° loading ramp with coefficient of friction =0.3. The safety pusher applies a 60 N force parallel to the ramp surface.
    (a) (3 points) Calculate the package's acceleration to ensure safe handling
    (b) (2 points) Verify the normal force stays within ramp's 100 N safety limit

Solution: [8cm]

24. **Playground Swing Physics:** A child (2 kg mass) on a 1.5 m long swing is pushed to 1.5 m height relative to the lowest point.
    (a) (2 points) Calculate maximum speed at the swing's lowest point
    (b) (3 points) Determine the rope tension at this point (safety threshold: 60 N)

Solution:

25. **Traffic Accident Investigation:** A compact car (1200 kg) traveling at 90 km/h (25 m/s) collides with a stationary delivery truck (1800 kg) at an intersection.
    (a) (3 points) Calculate their combined speed after collision if locked together
    (b) (2 points) Determine the energy converted to damage and heat

Solution:

26. **Satellite Deployment:** Engineers are positioning a weather satellite 500 km above Earth's surface.
    (a) (3 points) Calculate the gravitational acceleration at this altitude
    (b) (2 points) Compare to Earth's surface gravity

Solution:

27. **Aquarium Light Guide:** An underwater camera system uses glass (n=1.52) to water (n=1.33) light transmission at 40° entry angle.
    (a) (3 points) Calculate the light's path angle in water
    (b) (2 points) Determine the critical angle for total internal reflection

Solution: